

Proximal Drift-Plus-Penalty for Weakly Convex Optimization with Time-Average Constraints: A Soft Prompting Motivation

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Abstract

Soft prompting provides a concrete setting in which an optimization objective must be balanced against operational constraints that are naturally imposed over a long horizon. We consider a stationary prompt-selection policy and use this setting to motivate weakly convex optimization with one ordinary constraint and one long-run constraint that may itself change from round to round. We develop a proximal drift-plus-penalty (DPP) method in which the long-run constraint is stabilized before queue weighting. The stabilization makes the queue-weighted constraint component strongly convex and yields a proximal subproblem with a queue-independent baseline curvature. We prove a pointwise $O(V)$ virtual-queue bound, an $O(V/T)$ time-average constraint guarantee, and an $O(1/V + 1/T + V_T/T)$ bound on the squared near-stationarity residual of the penalized problem, where V_T is the cumulative functional variation of the per-round constraint loss. With additional Lipschitz and violating-point regularity assumptions, we convert this penalized stationarity guarantee into an $O(1/V + 1/T + V_T/T)$ expected violation of the ordinary constraint for a uniformly sampled iterate. The resulting statement is deliberately a nearly-KKT guarantee rather than a claim of convergence of the entire nonconvex iterate sequence to an exact KKT point.

1 Problem Formulation and Assumptions

We study the constrained problem

$$\min_{x \in \mathcal{X}} f(x) \quad \text{s.t.} \quad g(x) \leq 0, \quad \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T h_t(x_t) \leq 0, \quad (1)$$

where h_t is the constraint loss in force at round t : if x_t denotes the decision used at round t , the operational requirement is on the *time average* of the sequence $h_1, h_2, \dots$ evaluated along the algorithm's own trajectory. Nothing below assumes h_t is the same function from one round to the next.

We enforce the ordinary constraint through the exact hinge penalty

$$P(x) = f(x) + \beta[g(x)]_+, \quad [a]_+ := \max\{a, 0\}. \quad (2)$$

Assumption 1 (Decision set and boundedness). $\mathcal{X}$ is nonempty, closed, convex, and compact, with diameter $\sup_{x,y \in \mathcal{X}} \|x - y\| \leq D$. The functions f, g , and every h_t are finite on an open neighborhood of $\mathcal{X}$.

Assumption 2 (Weak convexity). f is ρ_f -weakly convex, g is ρ_g -weakly convex, and for every t , h_t is ρ_h -weakly convex on a neighborhood of $\mathcal{X}$. Since $[g]_+$ is also ρ_g -weakly convex, $\rho_P := \rho_f + \beta\rho_g$ is a valid weak-convexity modulus for P .

Assumption 3 (Lipschitz continuity). f, g , and every h_t are respectively L_f, L_g, L_h -Lipschitz on $\mathcal{X}$.

Assumption 4 (Bounded constraint values). For all t and all $x \in \mathcal{X}$, $|h_t(x)| \leq H$.

Assumption 5 (Strict time-average feasibility). There exists a point $x^\circ \in \mathcal{X}$ and $\xi > 0$ such that $h_t(x^\circ) \leq -\xi$ for all t .

This is a Slater-type condition for the time-average constraint, required to hold at a single common comparison point across every round. **Importantly**, x° need not minimize P — it is used only as a comparison point in the queue argument, and the proof of Lemma 2 below drops $P(x_{t+1})$ only as far as its global lower bound $P_{\min}$, never as far as $P(x^\circ)$ itself. The strict margin is needed because the stabilization term $\gamma\|x - x_t\|^2$ consumes part of it.

Assumption 6 (Violating-point regularity for the ordinary constraint). *There exists $\kappa_g > 0$ such that, for every $x \in \mathcal{X}$ satisfying $g(x) > 0$,*

$$\text{dist}(0, \beta \partial g(x) + N_{\mathcal{X}}(x)) \geq \kappa_g.$$

For a smooth constraint and an interior point of $\mathcal{X}$, this reduces to $\beta \|\nabla g(x)\| \geq \kappa_g$ whenever $g(x) > 0$.

Assumption 7 (Algorithmic parameters). *Choose*

$$\frac{\rho_h}{2} < \gamma < \frac{\xi}{D^2}, \quad 0 < \eta < \frac{1}{\rho_P}$$

(any $\eta > 0$ when $\rho_P = 0$), and let $V \geq 1$. Initialize $Q_1 = 0$.

2 Algorithm

At round t , only h_t — the constraint function *currently in force* — is observed and used; nothing about $h_{t+1}, h_{t+2}, \dots$ is required in advance.

Algorithm 1 Proximal Drift-Plus-Penalty

Require: Initial point $x_1 \in \mathcal{X}$, queue $Q_1 = 0$, parameters $V \geq 1, \eta > 0, \gamma > 0$

1: **for** $t = 1, \dots, T$ **do**

2: Observe the current round's constraint function h_t

3: Compute

$$x_{t+1} \in \underset{x \in \mathcal{X}}{\text{argmin}} \left\{ P(x) + \frac{Q_t}{V} [h_t(x) + \gamma \|x - x_t\|^2] + \frac{1}{2\eta} \|x - x_t\|^2 \right\}$$

4: Update the virtual queue

$$Q_{t+1} \leftarrow [Q_t + h_t(x_{t+1})]_+$$

5: **end for**

Lemma 1 (Uniform well-posedness). *Under Assumptions 1–7, the subproblem in Algorithm 1 is strongly convex with modulus*

$$\mu_t = \frac{1}{\eta} - \rho_P + q_t(2\gamma - \rho_h) \geq \mu_0 := \frac{1}{\eta} - \rho_P > 0, \quad q_t := \frac{Q_t}{V},$$

for every t , regardless of which h_t is in force that round. Consequently x_{t+1} is unique.

3 Pointwise Queue Bound

Let $P_{\min} := \min_{x \in \mathcal{X}} P(x)$, $P_{\max} := \max_{x \in \mathcal{X}} P(x)$, and define

$$C_{\circ} := P(x^{\circ}) - P_{\min} + \frac{D^2}{2\eta}, \quad \bar{\xi} := \xi - \gamma D^2 > 0, \quad \bar{q} := \frac{2C_{\circ}}{\bar{\xi}} + H.$$

Lemma 2 (Pointwise queue bound). *Under Assumptions 1–7, $Q_t \leq 2VC_{\circ}/\bar{\xi} + H$ for all t . Consequently, for $V \geq 1$, $q_t \leq \bar{q}$ for all t .*

This holds already under Assumptions 1–7 as stated — each h_t may differ from round to round, so long as it satisfies the *uniform* moduli/bound/margin those assumptions require of *every* t .

4 Functional Variation and the Movement/Stationarity Bounds

The queue bound above does not depend on how much h_t changes from one round to the next. The movement bound does: a lower bound on how h_{t+1} differs from h_t at the same point x_{t+1} appears when the per-round potential is telescoped, and it must be paid for explicitly.

Definition 1 (Functional variation). Define the one-step functional variation $\Delta_t := \sup_{x \in \mathcal{X}} |h_{t+1}(x) - h_t(x)|$ and the cumulative variation $V_T := \sum_{t=1}^{T-1} \Delta_t$; write $\nu_T := V_T/T$ for the average functional variation.

Assumption 8 (Uniformity across rounds). The moduli ρ_h, L_h, H in Assumptions 2–4 and the point x° and margin ξ in Assumption 5 are common to every t (not merely existing separately for each t).

Lemma 3 (Well-posedness and pointwise queue bound, restated). Under Assumption 8, Lemma 1 and Lemma 2 hold with the same constants $\mu_0, C_\circ, \bar{\xi}, \bar{q}$ at every t , uniformly over the whole run.

Theorem 1 (Long-run constraint feasibility). Under the assumptions above,

$$\frac{1}{T} \sum_{t=1}^T h_t(x_{t+1}) \leq \frac{2VC_\circ}{\bar{\xi}T} + \frac{H}{T}. \quad (3)$$

Consequently, the time-average constraint violation is $O(V/T)$.

Theorem 2 (Movement and nearly-stationary guarantee). Define $c_0 := 1/\eta - \rho_P/2 > 0$, $C_S := P_{\max} - P_{\min} + 2\bar{q}H$, $A := 1/\eta + 2\gamma\bar{q}$. Let τ be uniform on $\{1, \dots, T\}$, and define

$$R_t := \text{dist}(0, \partial P(x_{t+1}) + q_t \partial h_t(x_{t+1}) + N_{\mathcal{X}}(x_{t+1})).$$

Then

$$\frac{1}{T} \sum_{t=1}^T \|x_{t+1} - x_t\|^2 \leq \frac{C_S}{c_0 T} + \frac{H^2}{c_0 V} + \frac{\bar{q} V_T}{c_0 T} + \frac{D^2}{T}, \quad (4)$$

and

$$\mathbb{E}[R_\tau^2] \leq A^2 \cdot (\text{the right-hand side of (4)}) = O\left(\frac{1}{V} + \frac{1}{T} + \frac{V_T}{T}\right). \quad (5)$$

Consequently $\mathbb{E}[R_\tau] = O(V^{-1/2} + T^{-1/2} + \sqrt{V_T/T})$. If $h_t \equiv h$ is in fact constant in t , then $V_T = 0$ and this recovers the time-invariant movement bound exactly.

Theorem 3 (Nearly-feasible stationarity for the ordinary constraint). Suppose in addition that $\text{dist}(0, \beta \partial g(x) + N_{\mathcal{X}}(x)) \geq \kappa_g$ whenever $g(x) > 0$, that L_f, L_h are uniform subgradient bounds, and define $\delta_g := \kappa_g - L_f - \bar{q}L_h > 0$, $G_g := \sup_{x \in \mathcal{X}} [g(x)]_+$. Then

$$\mathbb{E}[g(x_{\tau+1})]_+ \leq \frac{G_g}{\delta_g^2} \mathbb{E}[R_\tau^2] = O\left(\frac{1}{V} + \frac{1}{T} + \frac{V_T}{T}\right). \quad (6)$$

Remark 1 (Interpretation). The temporal variation contributes only through V_T/T in the squared stationarity residual. Thus $V_T = o(T)$ is sufficient for the variation penalty to vanish, while $V_T = O(1)$ recovers the stationary-order bound. A constraint that changes by a bounded total amount over the whole horizon costs nothing asymptotically; one that keeps drifting at a constant per-round rate ($V_T = \Theta(T)$) leaves a floor in the movement and residual bounds that $T \rightarrow \infty$ alone cannot remove.

Note: proofs of Lemma 1, Lemma 2, and Theorems 1–3 were not part of the pasted text this draft was reconstructed from, so none are included here. A structurally equivalent time-invariant version of this paper (shown earlier in the same conversation) does contain full appendix proofs, built on the Lyapunov function $W_t = \frac{1}{2}Q_t^2 + \frac{V}{2\eta}\|x_t - x^\circ\|^2$; adapting them to h_t only requires tracking the extra term $q_{t+1}(h_{t+1}(x_{t+1}) - h_t(x_{t+1})) \leq \bar{q}\Delta_t$ that appears when the per-round potential is telescoped across a round where the constraint function itself changed.

$$f(x) = \frac{1}{2}(0.2)|x|^2 - c_f^\top x + \sum_{j=1}^6 0.2 \cos(w_j^f \cdot x + b_j^f)$$

$$g(x) = |x|^2 - r_g^2, \quad r_g = 0.75$$

$$h_t(x) = \underbrace{\frac{1}{2}(2.0)|x|^2 + \sum_{j=1}^3 0.025 \cos(w_j^h \cdot x + b_j^h)}_{\text{fixed; } +; \underbrace{\psi_t}_{\text{changes each round}}} - 0.400722 h(x),$$

So the full subproblem, with everything substituted in: